

Question-2: Implement a circuit to convert the square wave having $f=500\text{Hz}$, and input voltage $V_{pp}=2\text{V}$ to triangular waveform. Draw the complete circuit. Also draw the labelled input and the output waveforms in the box given below.

Hint: Use $R_1=33\text{k}\Omega$, $R_f=100\text{k}\Omega$ and $C=33\text{nF}$.

[Marks 15] CLO1

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	EL224-Electronic Circuit Design (Lab)	Course Code:	EL224
	Program:	BS Electrical Engineering	Semester:	Fall 2018
	Duration:	60 Minutes	Total Marks:	30
	Date:	11 Dec, 2018	Weight	18
	Section:	EL-B	Page(s):	2
	Exam Type:	Final Paper (Practical)		

Student : Name: _____ Roll No. _____ Section: _____

- Instruction/Notes:
1. Read the Questions carefully.
 2. Write all Calculations on this paper.
 3. Draw all output waveforms for each experiment.

Question-1: Design and implement a function generator using 555 timer in Astable configuration. The design parameters are given below:

Frequency	Duty cycle	Choices of Capacitances for Astable MVB
1kHz	80%	1nF, 4.7nF, 10nF, 33nF, 47nF

Show your calculations for the resistances using the circuit diagram shown in figure-1. For practical implementation, show all the waveforms on the oscilloscope. Note down the observed frequency, duty cycle and peak to peak values of output waveforms. Also find the percentage error.
(You are allowed a tolerance of $\pm 15\%$ between practical and designed values.) [Marks 15] CLO3

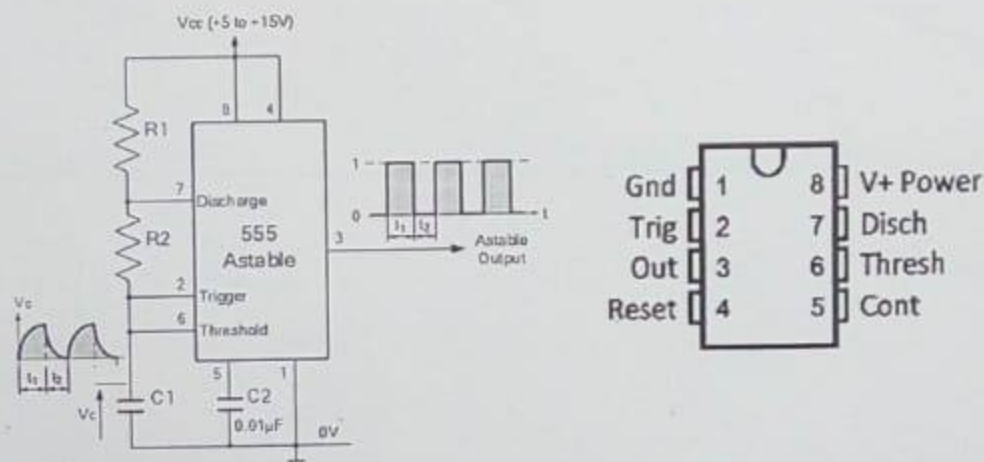


Figure 1- Astable Multivibrator using 555 Timer

Calculated values:

Ra=_____, Rb=_____, C=_____, Frequency=_____, Duty Cycle=_____.